

Exno : 08

Rollno: 241001044

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Backward Chaining

Exercise 1: Medical Diagnosis

% Facts

has_fever(john).

has_cough(john).

% Rule

has_flu(X) :- has_fever(X), has_cough(X).

% Query

% ?- has_flu(john).

Output:

true.

Exercise 2: Student Eligibility Checker

% Facts

passed(alice, maths).

passed(alice, science).

% Rule

eligible(X) :- passed(X, maths), passed(X, science).

% Query

% ?- eligible(alice).

Output:

true.

Exercise 3: Travel Planner

% Facts

has_ticket(james).

```
has_visa(james).
```

```
% Rule
```

```
can_travel(X) :- has_ticket(X), has_visa(X).
```

```
% Query
```

```
% ?- can_travel(james).
```

Output:

```
true.
```

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```
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```

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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
18 ?- consult(['C:/Users/Lenovo/Documents/expno 8.pl']).
true.
```

```
19 ?- consult(['C:/Users/Lenovo/Documents/expno 9.pl']).
true.
```

```
20 ?- consult(['C:/Users/Lenovo/Documents/expno 10.pl']).
true.
```

```
21 ?- can_travel(james).
true.
```

```
22 ?- eligible(alice).
true.
```

```
23 ?- has_flu(john).
true.
```

```
24 ?- ◆
```